CSE-476: Introduction to Natural Language Processing
Quiz 1

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

Q1. In the BIO tagging scheme for NER, what does the "B-" prefix indicate?

(A) Between two tokens
(B) Beginning of an entity
(C) Inside an entity
(D) Outside any entity

Q2. Which statement best describes a (plain) Markov chain in sequence tagging?

(A) It uses only words (observations) and ignores labels
(B) It models transitions between labels but does not consider the words
(C) It models emissions P(word | label)
(D) It is identical to an HMM

Q3. In a Hidden Markov Model (HMM) for NER, an emission probability is:

(A) P(label | previous label)
(B) P(word | label)
(C) P(label | word)
(D) P(word | previous word)

Q4. In an n-gram language model, a bigram LM predicts the next word using:

(A) No previous words
(B) The previous 1 word
(C) The previous 2 words
(D) The entire sentence history

Q5. Which statement about perplexity is correct?

(A) It's the raw probability of a sentence

(B) It's the inverse probability of a test text, normalized by length
(C) It's the fraction of correctly labeled tokens
(D) It's the number of unique words in the corpus

Q6. Which of the following best describes the main goal of tokenization in NLP?

(A) To label each token with a grammatical category
(B) To convert text into a sequence of symbols suitable for modeling
(C) To reduce vocabulary size to zero
(D) To assign probabilities to sentences

Q7. Why is splitting text by spaces (text.split(" ")) considered an insufficient tokenizer?

(A) It produces too many tokens
(B) It does not work well for languages without explicit word boundaries
(C) It cannot tokenize English at all
(D) It ignores punctuation entirely

Q8. Which of the following is NOT listed as a common sequence tagging task?

(A) Part-of-speech tagging
(B) Named Entity Recognition
(C) Dependency parsing
(D) Sentiment analysis

Q9. What is a common limitation of BIO tagging for Named Entity Recognition?

(A) It cannot represent overlapping or nested entities
(B) It requires too many labels
(C) It does not support multiple entity types
(D) It ignores entity boundaries

Q10. When choosing between character-level tokenization and sentence-level tokenization, which trade-off must be considered?

(A) Speed vs accuracy
(B) Tokenized sequence length vs vocabulary size
(C) Precision vs recall
(D) Memory vs computation

End of Paper